SmartQueue: Development of a Technology-Enabled Digital Queue Management System to Minimize Patient Waiting Times and Improve the Efficiency of Hospital Services

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I. INTRODUCTION

Hospitals play an important role in providing patients with consultations, diagnostic procedures, treatments, medications, and other essential healthcare services. However, one of the common challenges encountered in hospitals is the long waiting time experienced by patients before receiving healthcare services. Waiting may occur during registration, consultation, laboratory examinations, diagnostic procedures, payment, and pharmacy services. When the number of patients exceeds the available capacity of healthcare facilities, delays may occur and affect both patients and healthcare workers.

According to Safitri et al. (2024), waiting time is an important factor associated with outpatient satisfaction in hospitals. Their meta-analysis found that patients who experienced shorter waiting times of less than two hours had significantly greater odds of being satisfied compared with patients who experienced longer waiting times. This demonstrates that waiting time is not simply an inconvenience but an important component of the overall patient experience (Safitri et al., 2024).

The problem is also evident in the Philippine healthcare system. According to Conda et al. (2025), only 35.5% of hospitals included in their baseline assessment were able to schedule medical appointments within two weeks, while 68.5% of hospitals accepted walk-in patients for medical consultations. The same assessment reported that only 40.9% of hospitals could schedule laboratory tests within two weeks, demonstrating differences in scheduling efficiency and access to healthcare services (Conda et al., 2025).

Furthermore, Manderson et al. (2025) explained that increasing demand for healthcare services, workforce limitations, resource constraints, and inefficiencies in service delivery can contribute to long waiting times in outpatient and community healthcare settings. Their systematic review found that service-level interventions can reduce waiting times and waiting lists, although the certainty of evidence varies across interventions (Manderson et al., 2025).

In this proposal, the author seeks to address the problem of long hospital waiting times by proposing SmartQueue, an information technology-enabled digital queue management system. The proposed system aims to organize patient queues, provide real-time queue information, improve communication between patients and hospital staff, and support more efficient patient flow.

II. PROBLEM DESCRIPTION

Long waiting times in hospitals refer to the excessive amount of time patients spend waiting before receiving healthcare services. These delays may occur at different stages of a patient's hospital visit, including registration, consultation, laboratory testing, diagnostic procedures, payment, and medication collection. When patients are required to wait in several separate queues without sufficient information regarding their expected waiting time, congestion and uncertainty may increase.

According to Conda et al. (2025), hospital scheduling and service efficiency vary considerably across healthcare facilities in the Philippines. Their assessment found that 68.5% of hospitals accepted walk-in patients, while only 35.5% were able to schedule medical appointments within two weeks. Laboratory scheduling also showed limitations, with only 40.9% of hospitals able to schedule laboratory tests within two weeks (Conda et al., 2025).

One factor contributing to waiting times is the imbalance between patient demand and available healthcare resources. Manderson et al. (2025) noted that increasing healthcare demand, an aging population, chronic health conditions, reduced workforce capacity, and resource constraints can contribute to longer waiting times. Inefficient referral, booking, and patient-flow processes may further increase delays (Manderson et al., 2025).

Long waiting times can also influence how patients perceive the quality of healthcare services. According to Safitri et al. (2024), patients who experienced shorter waiting times were significantly more likely to report satisfaction with outpatient services. Their meta-analysis showed that patients who waited less than two hours had 6.86 times greater odds of being satisfied compared with those who waited longer than two hours (Safitri et al., 2024).

The effects of waiting are not limited to patient satisfaction. Manderson et al. (2025) reported that consumers waiting for healthcare services described delays as affecting their health outcomes and causing distress and anxiety. The review also found that 18 of the 22 included studies reported sustained reductions in waiting time or waiting lists following service-level interventions, although the overall certainty of evidence was low to very low (Manderson et al., 2025).

In addition, long waiting times may create challenges for healthcare workers and hospital administrators. When queues become congested, healthcare workers may have difficulty maintaining an organized patient flow while simultaneously performing administrative and clinical responsibilities. This highlights the need for systems that can provide hospitals with better information regarding patient volume, queue status, and service demand.

With these challenges in mind, the use of information technology may provide an opportunity to improve the organization of patient queues and communication between patients and healthcare providers.

III. PROPOSAL DESCRIPTION

The proposed solution is the development of SmartQueue, an information technology-enabled digital queue management system designed to improve patient flow and reduce unnecessary waiting in hospitals. The system would allow patients to digitally register for hospital services and receive a queue number without having to remain in a physical line.

According to Dutta et al. (2023), a Smart Queuing Management System for digital healthcare can improve the organization of patients by using a digital system to consider factors such as appointment time, age, and disease severity. The authors proposed that a technology-based queueing system can help organize patients, reduce waiting times, and support timely healthcare delivery.

SmartQueue would allow patients to register through a hospital kiosk, QR code, website, or mobile application. After registration, the system would provide a digital queue number and estimated waiting time. Patients could monitor their queue status through their mobile device or a digital display within the hospital. Automated notifications could also inform patients when their queue number is approaching, reducing the need to remain continuously inside a crowded waiting area.

The system would also provide hospital staff with a centralized dashboard where they could monitor the number of waiting patients, patients currently being served, queue status, and estimated waiting times. By having access to real-time information, healthcare workers and administrators could identify areas experiencing congestion and make appropriate adjustments to patient flow.

The proposed SmartQueue system may include the following features:

1. Digital Registration – Patients can register through a kiosk, QR code, website, or mobile application.
2. Digital Queue Number – Each patient receives a unique queue number.

3. Real-Time Queue Tracking – Patients can monitor their current position in the queue.
4. Estimated Waiting Time – The system provides an estimated time before the patient's turn.
5. Automated Notifications – Patients receive alerts when their queue number is approaching.
6. Hospital Staff Dashboard – Healthcare personnel can monitor patient volume and queue status.
7. Priority Management – Hospital-approved priority categories can be incorporated while maintaining professional clinical triage.
8. Data and Reports – The system can generate reports regarding patient volume, peak hours, average waiting times, and service utilization.

According to Manderson et al. (2025), interventions that modify referral processes, booking systems, and models of care can contribute to sustained reductions in healthcare waiting times. Their findings support the use of organized service-level approaches to improve patient flow, although further high-quality research is still needed to determine which interventions provide the greatest benefit in specific healthcare settings (Manderson et al., 2025).

The primary users of SmartQueue would be hospital outpatients, healthcare workers, and hospital administrators. Patients would benefit from having access to real-time queue information, while healthcare workers could use the system to better organize patient flow. Hospital administrators could use the collected data to identify bottlenecks, evaluate service efficiency, and improve scheduling decisions.

However, SmartQueue would not replace clinical judgment or emergency triage. Patients requiring immediate medical attention would continue to be assessed and prioritized by qualified healthcare professionals according to hospital policies and clinical protocols. The expected impact of SmartQueue includes improved patient-flow management, reduced unnecessary waiting, decreased congestion in waiting areas, improved communication, and increased patient satisfaction. Since recent evidence indicates that service-level interventions can reduce waiting times, a digital queue management system may serve as one component of a broader strategy for improving hospital efficiency (Manderson et al., 2025).

IV. CONCLUSION

In conclusion, long waiting times in hospitals are an important problem because they can affect patients' satisfaction, comfort, and overall healthcare experience. Patients may spend considerable time waiting for registration, consultation, laboratory examinations, medications, and other healthcare services. Based on the reviewed studies, improving the organization of hospital services and patient flow can help minimize unnecessary delays.

To address this problem, the proposed SmartQueue system provides a technology-based approach to improving hospital queue management. Through digital registration, real-time queue tracking, estimated waiting times, and automated notifications, the system can help patients manage their waiting time more conveniently. It can also assist healthcare workers and hospital administrators in monitoring patient flow and identifying areas where delays commonly occur.

Overall, SmartQueue has the potential to make hospital services more organized, efficient, and patient-centered. Although its implementation would require proper planning, resources, and cooperation from hospital staff, integrating information technology into queue management can provide a practical approach to reducing unnecessary waiting times and improving the overall healthcare experience.

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